MAT 2040 Linear Algebra

Assignment 2

- Released date: 2024/06/11.
- Due: 2024/06/18.
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW2".

Question 1. The matrix

$$A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

has the property that $A^2 = O$. Is it possible for a nonzero symmetric 2×2 matrix to have this property? Prove your answer.

Solution

Impossible.

Suppose a 2×2 symmetric matrix is one of the form

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

Thus

$$A^2 = \begin{pmatrix} a^2 + b^2 & ab + bc \\ ab + bc & b^2 + c^2 \end{pmatrix}$$

If $A^2 = O$, then its diagonal entries must be 0 .

$$a^2 + b^2 = 0 \quad \text{and} \quad b^2 + c^2 = 0$$

Thus $a = b = c = 0$ and hence $A = O$.

Question 2 . Let C be a nonsymmetric $n \times n$ matrix. For each of the following, determine whether the given matrix must necessarily be symmetric or could possibly be nonsymmetric:

(a) $A = C + C^T$
(b) $B = C - C^T$
(c) $D = C^T C$
(d) $E = C^T C - C C^T$
(e) $F = (I + C)\left(I + C^T\right)$
(f) $G = (I + C)\left(I - C^T\right)$

Solution

(a) The matrix A is symmetric since

$$A^T = \left(C + C^T\right)^T = C^T + \left(C^T\right)^T = C^T + C = A$$

(b) The matrix B is not symmetric since

$$B^T = \left(C - C^T\right)^T = C^T - \left(C^T\right)^T = C^T - C = -B$$

(c) The matrix D is symmetric since

$$A^T = \left(C^T C\right)^T = C^T \left(C^T\right)^T = C^T C = D$$

(d) The matrix E is symmetric since

$$\begin{aligned} E^T &= \left(C^T C - C C^T\right)^T = \left(C^T C\right)^T - \left(C C^T\right)^T \\ &= C^T \left(C^T\right)^T - \left(C^T\right)^T C^T = C^T C - C C^T = E \end{aligned}$$

(e) The matrix F is symmetric since

$$F^T = \left((I + C)\left(I + C^T\right)\right)^T = \left(I + C^T\right)^T (I + C)^T = (I + C)\left(I + C^T\right) = F$$

(e) The matrix G is not symmetric.

$$\begin{aligned} F &= (I + C)\left(I - C^T\right) = I + C - C^T - C C^T \\ F^T &= \left((I + C)\left(I - C^T\right)\right)^T = \left(I - C^T\right)^T (I + C)^T \\ &= (I - C)\left(I + C^T\right) = I - C + C^T - C C^T \end{aligned}$$

F and F^T are not the same. The two middle terms $C - C^T$ and $-C + C^T$ do not agree.

Question 3. Let

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Show that if $d = a_{11}a_{22} - a_{21}a_{12} \neq 0$, then

$$A^{-1} = \frac{1}{d}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Solution

If $d = a_{11}a_{22} - a_{21}a_{12} \neq 0$, then

$$\frac{1}{d}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \frac{a_{11}a_{22}-a_{12}a_{21}}{d} & 0 \\ 0 & \frac{a_{11}a_{22}-a_{12}a_{21}}{d} \end{pmatrix} = I$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\left[\frac{1}{d}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}\right] = \begin{pmatrix} \frac{a_{11}a_{22}-a_{12}a_{21}}{d} & 0 \\ 0 & \frac{a_{11}a_{22}-a_{12}a_{21}}{d} \end{pmatrix} = I$$

Therefore

$$\frac{1}{d}\begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix} = A^{-1}$$

Question 4 . Let A and B are $n \times n$ matrices. Show that if

$$AB = A \text{ and } B \neq I$$

then A must be singular.

Solution

If A were nonsingular and $AB = A$, then it would follow that $A^{-1}AB = A^{-1}A$ and hence that $B = I$. So if $B \neq I$, then A must be singular.

Question 5. Prove that if A is nonsingular then A^T is nonsingular and

$$\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$$

Hint: $(AB)^T = B^T A^T$.

Solution

Since

$$A^T\left(A^{-1}\right)^T = \left(A^{-1}A\right)^T = I$$
$$\left(A^{-1}\right)^T A^T = \left(AA^{-1}\right)^T = I$$

it follows that

$$\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$$

Question 6. Let A be an $m \times n$ matrix. Show that A^TA and AA^T are both symmetric.

Solution

$$\left(A^TA\right)^T = A^T\left(A^T\right)^T = A^TA$$
$$\left(AA^T\right)^T = \left(A^T\right)^TA^T = AA^T$$

Question 7. Let

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 2 & 2 & 6 \end{pmatrix},$$
$$C = \begin{pmatrix} 1 & 2 & 4 \\ 0 & -1 & -3 \\ 2 & 2 & 6 \end{pmatrix}$$

(a) Find an elementary matrix E such that $EA = B$

(b) Find an elementary matrix F such that $FB = C$.

(c) Is C row equivalent to A ? Explain.

Solution

(a) $E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

(b) $E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$

(c) Since

$$C = FB = FEA$$

where F and E are elementary matrices, it follows that C is row equivalent to A.

Question 8. Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 6 & 4 & 5 \\ 4 & 1 & 3 \end{pmatrix}$$

(a) Find elementary matrices E_1, E_2, E_3 such that

$$E_3E_2E_1A = U$$

where U is an upper triangular matrix.
(b) Determine the inverses of E_1, E_2, E_3 and set $L = E_1^{-1}E_2^{-1}E_3^{-1}$. What type of matrix is L ? Verify that $A = LU$.

Solution

(1)$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

(b) $E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $E_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$, $E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$

The product $L = E_1^{-1}E_2^{-1}E_3^{-1}$ is lower triangular.

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

Question 9. Let

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 4 \end{pmatrix}$$

(a) Express A^{-1} as a product of elementary matrices.
(b) Express A as a product of elementary matrices.

Solution

A can be reduced to the identity matrix using three row operations

$$\begin{pmatrix} 2 & 1 \\ 6 & 4 \end{pmatrix} \to \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \to \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The elementary matrices corresponding to the three row operations are

$$E_1 = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$$

So

$$E_3E_2E_1A = I$$

and hence

$$A = (E_3E_2E_1)^{-1} = E_1^{-1}E_3^{-1}E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

and $A^{-1} = E_3E_2E_1$.

Question 10. Compute the LU factorization of each of the following matrices.

(a) $\begin{bmatrix} 3 & 1 \\ 9 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix}$

(d) $\begin{bmatrix} -2 & 1 & 2 \\ 4 & 1 & -2 \\ -6 & -3 & 4 \end{bmatrix}$

Solution

- **part (a)**

$$\begin{bmatrix} 3 & 1 \\ 9 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

- **part (b)**

$$\begin{bmatrix} 2 & 4 \\ -2 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix} \quad L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix}$$

- **part (c)**

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 6 \\ -2 & 2 & 7 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 4 & 9 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

- **part (d)**

$$\begin{bmatrix} -2 & 1 & 2 \\ 4 & 1 & -2 \\ -6 & -3 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & -6 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}, U = \begin{bmatrix} -2 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

Question 11. Find the inverse of each of the following matrics.

(a) $\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & 6 \\ 3 & 8 \end{bmatrix}$

(d) $\begin{bmatrix} 3 & 0 \\ 9 & 3 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(f) $\begin{bmatrix} 2 & 0 & 5 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

(g) $\begin{bmatrix} -1 & -3 & -3 \\ 2 & 6 & 1 \\ 3 & 8 & 3 \end{bmatrix}$

(h) $\begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix}$

Solution

- **part (a)**

$$\left[\begin{array}{cc|cc} -1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & -1 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{array}\right]$$

  Its inverse is $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

- **part (b)** $\begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$
- **part (c)** $\begin{bmatrix} -4 & 3 \\ \frac{3}{2} & -1 \end{bmatrix}$
- **part (d)** $\begin{bmatrix} \frac{1}{3} & 0 \\ -1 & \frac{1}{3} \end{bmatrix}$
- **part (e)** $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$
- **part (f)** $\begin{bmatrix} 3 & 0 & -5 \\ 0 & \frac{1}{3} & 0 \\ -1 & 0 & 2 \end{bmatrix}$
- **part (g)** $\begin{bmatrix} 2 & -3 & 3 \\ -0.6 & 1.2 & -1 \\ -0.4 & -0.2 & 0 \end{bmatrix}$
- **part (h)** $\begin{bmatrix} -0.5 & -1 & -0.5 \\ -2 & -1 & -1 \\ 1.5 & 1 & 0.5 \end{bmatrix}$

Question 12. Let A be a 3×3 matrix and suppose that

$$2\mathbf{a}_1 + \mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}$$

How many solutions will the system $A\mathbf{x} = \mathbf{0}$ have? Explain. Is A nonsingular? Explain.

Solution

If we set $\mathbf{x} = (2, 1, -4)^T$, then

$$A\mathbf{x} = 2\mathbf{a}_1 + 1\mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}$$

Thus $\mathbf{x}$ is a nonzero solution to the system $A\mathbf{x} = \mathbf{0}$. But if a homogeneous system has a nonzero solution, then it must have infinitely many solutions. In particular, if c is any scalar, then $c\mathbf{x}$ is also a solution to the system since

$$A(c\mathbf{x}) = cA\mathbf{x} = c\mathbf{0} = \mathbf{0}$$

Since $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$ it follows that the matrix A must be singular.

Question 13. Show that if A is a symmetric nonsingular matrix then A^{-1} is also symmetric.

Solution

If A is symmetric and nonsingular, then

$$(A^{-1})^T = (A^{-1})^T(AA^{-1}) = (A^TA^{-1})^TA^{-1} = (AA^{-1})^TA^{-1} = A^{-1}$$

Question 14. Prove that B is row equivalent to A if and only if there exists a nonsingular matrix M such that $B = MA$.

Solution

If B is row equivalent to A, then there exist elementary matrices $E_1, E_2, \ldots, E_k$ such that

$$B = E_kE_{k-1}\cdots E_1A$$

Let $M = E_kE_{k-1}\cdots E_1$. The matrix M is nonsingular since each of the E_i's is nonsingular.

Conversely suppose there exists a nonsingular matrix M such that $B = MA$. Since M is nonsingular it is row equivalent to I. Thus there exist elementary matrices $E_1, E_2, \ldots, E_k$ such that

$$M = E_kE_{k-1}\cdots E_1I$$

It follows that

$$B = MA = E_kE_{k-1}\cdots E_1A$$

Therefore B is row equivalent to A.

Question 15. Perform each of the following block multiplications:

(a) $\left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right] \left[\begin{array}{ccc} 4 & -2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \\ \hline 1 & 2 & 3 \end{array}\right]$

(b) $\left[\begin{array}{cc} 4 & -2 \\ 2 & 3 \\ 1 & 1 \\ \hline 1 & 2 \end{array}\right] \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right]$

(c) $\left[\begin{array}{cc|cc} \frac{3}{5} & -\frac{4}{5} & 0 & 0 \\ \frac{4}{5} & \frac{3}{5} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc|c} \frac{3}{5} & \frac{4}{5} & 0 \\ -\frac{4}{5} & \frac{3}{5} & 0 \\ \hline 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right]$

(d) $\left[\begin{array}{ccc|cc} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc} 1 & -1 \\ 2 & -2 \\ 3 & -3 \\ \hline 4 & -4 \\ 5 & -5 \end{array}\right]$

Solution

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 4 & -2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 1 \\ 11 & -1 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & -2 \\ 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 8 & 5 & 8 \\ 3 & 2 & 3 \end{bmatrix} \quad \begin{bmatrix} 4 & -2 \\ 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -5 \\ -2 \end{bmatrix}$

$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 5 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = -3$

$\left[\begin{array}{cc} 4 & -2 \\ 2 & 3 \\ 1 & 1 \\ \hline 1 & 2 \end{array}\right] \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 2 & 1 & 2 & -1 \end{array}\right] = \begin{bmatrix} 0 & 2 & 0 & -2 \\ 8 & 5 & 8 & -5 \\ 3 & 2 & 3 & -2 \\ 5 & 3 & 5 & -3 \end{bmatrix}$

(c) Let

$$A_{11} = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} \; A_{12} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$
$$A_{21} = \begin{bmatrix} 0 & 0 \end{bmatrix} \; A_{22} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

The block multiplication is performed as follows:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} A_{11}^T & A_{21}^T \\ A_{12}^T & A_{22}^T \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}A_{11}^T + A_{12}A_{12}^T & A_{11}A_{21}^T + A_{12}A_{22}^T \\ A_{21}A_{11}^T + A_{22}A_{12}^T & A_{21}A_{21}^T + A_{22}A_{22}^T \end{bmatrix} = \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]$$

(d)
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 4 & -4 \\ 5 & -5 \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 2 & -2 \\ 1 & -1 \end{bmatrix}$$
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & -4 \\ 5 & -5 \end{bmatrix} = \begin{bmatrix} 5 & -5 \\ 4 & -4 \end{bmatrix}$$
$$\left[\begin{array}{ccc|cc} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \left[\begin{array}{cc} 1 & -1 \\ 2 & -2 \\ 3 & -3 \\ \hline 4 & -4 \\ 5 & -5 \end{array}\right] = \begin{bmatrix} 3 & -3 \\ 2 & -2 \\ 1 & -1 \\ 5 & -5 \\ 4 & -4 \end{bmatrix}$$

Question 16. Let A be an $m \times n$ matrix, X an $n \times r$ matrix, and B an $m \times r$ matrix. Show that

$$AX = B$$

if and only if

$$A\mathbf{x}_j = \mathbf{b}_j, j = 1, \ldots, r$$

Solution

$$AX = A(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_r) = (A\mathbf{x}_1, A\mathbf{x}_2, \ldots, A\mathbf{x}_r)$$

$$B = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r)$$

$AX = B$ if and only if the column vectors of AX and B are equal

$$A\mathbf{x}_j = \mathbf{b}_j \quad j = 1, \ldots, r$$

Question 17. Let

$$A = \begin{pmatrix} A_{11} & A_{12} \\ \mathbf{O} & A_{22} \end{pmatrix}$$

where all four blocks are $n \times n$ matrics.

(a) If A_{11} and A_{22} are nonsingular, show that A must also be nonsingular and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline \mathbf{O} & A_{22}^{-1} \end{array}\right]$$

(b) Determine C.

Solution

(1)

$$\begin{bmatrix} A_{11}^{-1} & C \\ \mathbf{O} & A_{22}^{-1} \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ \mathbf{O} & A_{22} \end{bmatrix} = \begin{bmatrix} I & A_{11}^{-1}A_{12} + CA_{22} \\ \mathbf{O} & I \end{bmatrix}$$

If

$$A_{11}^{-1}A_{12} + CA_{22} = \mathbf{O}$$

then

$$C = -A_{11}^{-1}A_{12}A_{22}^{-1}$$

Let

$$B = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ \mathbf{O} & A_{22}^{-1} \end{bmatrix}$$

Since $AB = BA = I$, it follows that $B = A^{-1}$.

(2)

$$C = -A_{11}^{-1}A_{12}A_{22}^{-1}$$